1 Correct common denominator of 15 with unsimplified numerator $3(2x) - 5(x-1)$ **[M1]**

$\frac{x+5}{15}$ **[A1]**

[Total Marks: 2]

2 M1 for a correct common denominator with at least one correct numerator, e.g.,
$\frac{5(2x-1)+3(x+4)}{15}$ or $\frac{10x-5}{15} + \frac{3x+12}{15}$

$\frac{13x+7}{15}$ **[A1]**

[Total Marks: 2]

3 $\frac{5(3x)-4(x-2)}{20}$ or $\frac{15x}{20} - \frac{4x-8}{20}$ **[M1]**

$\frac{11x+8}{20}$ **[A1]**

[Total Marks: 2]

4 Uses common denominator $(x+2)(3x-1)$ **[B1]**

Correct unsimplified numerator $5(3x-1)+2(x+2)$ **[M1]**

$\frac{17x-1}{(x+2)(3x-1)}$ or $\frac{17x-1}{3x^2+5x-2}$ **[A1]**

[Total Marks: 3]

5 Writes fractions over a common denominator of 20 **[M1]**

Correct unsimplified numerator: $5(x+3)-4(2x-1)$ or $5x+15-8x+4$ **[M1]**

$\frac{19-3x}{20}$ or $\frac{-3x+19}{20}$ **[A1]**

[Total Marks: 3]

6 Common denominator of 12 and correct numerator formulation: $\frac{3(3x)-4(x-2)}{12}$ **[M1]**

Numerator expanded correctly to $9x-4x+8$ **[M1]**

$\frac{5x+8}{12}$ **[A1]**

[Total Marks: 3]

7 $\frac{5(2x)-3(x-4)}{15}$ or $\frac{10x}{15} - \frac{3(x-4)}{15}$ **[M1]**

$10x-3x+12$ or better as numerator (allow one sign error for M1, e.g., $10x-3x-12$) **[M1]**

$\dfrac{7x+12}{15}$ **[A1]**

[Total Marks: 3]

8 **(a)** Uses a common denominator of 12, e.g. $\dfrac{4(2x)+3(x-1)}{12}$ **[M1]**

$\dfrac{11x-3}{12}$ **[A1]**

(b) $3x^2$ or $\dfrac{1}{y^2}$ seen **[M1]**

$\dfrac{3x^2}{y^2}$ or $3x^2y^{-2}$ **[A1]**

[Total Marks: 4]

9 **(a)** Common denominator 20 with numerator $5(3x) - 4 \times 2(x-3)$ or $15x - 8x + 24$ **[M1]**

$\dfrac{7x+24}{20}$ **[A1]**

(b) $\dfrac{6a^3b}{5c^2} \times \dfrac{10bc}{9a^2}$ **[M1]**

$\dfrac{4ab^2}{3c}$ **[A1]**

[Total Marks: 4]

10 **(a)** $\dfrac{15x+8x}{20}$ **[M1]**

$\dfrac{23x}{20}$ **[A1]**

(b) $\dfrac{5a^2b}{12} \times \dfrac{9}{10ab^2}$ **[M1]**

$\dfrac{3a}{8b}$ **[A1]**

[Total Marks: 4]

11 $(x-4)(x+4)$ seen **[B1]**

$(x-4)(x-1)$ seen **[B1]**

$\dfrac{x+4}{x-1}$ **[A1]**

[Total Marks: 3]

12 $3(x^2-4)$ or $3(x-2)(x+2)$ **[B1]**

$(x+5)(x-2)$ [B2] (or [M1] for $(x+a)(x+b)$ where $ab = -10$ or $a+b=3$)

$\frac{3(x+2)}{x+5}$ or $\frac{3x+6}{x+5}$ **[A1]**

[Total Marks: 4]

13 (a) $\frac{3(x+5)-2(x-2)}{(x-2)(x+5)}$ **[M1]**

$3x+15-2x+4$ correctly expanded on numerator **[M1]**

$\frac{x+19}{(x-2)(x+5)}$ or $\frac{x+19}{x^2+3x-10}$ **[A1]**

(b) $(x-4)(x+4)$ seen for numerator **[M1]**

$(x-4)(x+2)$ seen for denominator **[M1]**

$\frac{x+4}{x+2}$ **[A1]**

[Total Marks: 6]

14 (a) M1 for factorising the numerator correctly: $(x-3)(x+3)$

M1 for factorising the denominator correctly: $(2x-1)(x+3)$

$\frac{x-3}{2x-1}$ **[A1]**

(b) M1 for the correct common denominator $(x-2)(x+1)$

M1 for the correct unsimplified numerator $4(x+1)-3(x-2)$

$\frac{x+10}{(x-2)(x+1)}$ or $\frac{x+10}{x^2-x-2}$ **[A1]**

[Total Marks: 6]

15 (a) Common denominator $(x-3)(x+2)$ seen **[M1]**

Numerator $4(x+2)-3(x-3)$ or $4x+8-3x+9$ **[M1]**

$\frac{x+17}{(x-3)(x+2)}$ or $\frac{x+17}{x^2-x-6}$ **[A1]**

(b) Numerator factorised as $(x-3)(x+3)$ **[B1]**

Denominator factorised as $(x+2)(x+3)$ **[B1]**

$\frac{x-3}{x+2}$ **[B1]**

[Total Marks: 6]

16 Factorises denominator to $(2x-3)(2x+3)$ **[B1]**

Attempts to factorise numerator to $(2x+a)(x+b)$ where $ab=-12$ or $2b+a=5$ **[M1]**

Correctly factorises numerator to $(2x-3)(x+4)$ **[A1]**

$\frac{x+4}{2x+3}$ **[A1]**

[Total Marks: 4]

17 $(3x+1)(x-2)$ **[B2]**

If not B2, then B1 for $(3x+a)(x+b)$ where $ab=-2$ or $3b+a=-5$ **[B1]**

$(x-2)(x+2)$ **[B1]**

$(x-2)(x+1)$ **[B1]**

$\dfrac{3x^2-5x-2}{x^2-4}\times\dfrac{x^2-x-2}{3x+1}$ or equivalent correct inverted product **[M1]**

$\dfrac{(x-2)(x+1)}{x+2}$ or $\dfrac{x^2-x-2}{x+2}$ **[A1]**

[Total Marks: 6]

18 (a) Factorises $2x^2-5x-3=(2x+1)(x-3)$ and $4x^2-1=(2x-1)(2x+1)$ **[B1]**

Factorises $x^2-9=(x-3)(x+3)$ and $2x^2+5x-3=(2x-1)(x+3)$ **[B1]**

Inverts second fraction and multiplies: $\dfrac{(2x+1)(x-3)}{(2x-1)(2x+1)}\times\dfrac{(2x-1)(x+3)}{(x-3)(x+3)}$ **[M1]**

1 **[A1]**

(b) Factorises denominators: $(x-2)(x+2)$ and $(x+1)(x+2)$ **[M1]**

Common denominator $(x-2)(x+2)(x+1)$ with numerator $3(x+1)-2(x-2)$ **[M1]**

$\dfrac{x+7}{(x-2)(x+2)(x+1)}$ **[A1]**

[Total Marks: 7]

19 (a) Factorising numerator: $(2x+1)(x-3)$ **[B1]**

Factorising denominator: $(2x-3)(2x+3)$ **[B1]**

Note: no common factor cancels; the expression is already in its simplest form

Final answer: $\dfrac{2x^2-5x-3}{4x^2-9}$ or $\dfrac{(2x+1)(x-3)}{(2x-3)(2x+3)}$ **[A1]**

(b) Recognising $x^2-x-2=(x-2)(x+1)$ **[M1]**

Writing all over a common denominator of $(x-2)(x+1)$: $\dfrac{3(x+1)+4(x-2)-2x}{(x-2)(x+1)}$ **[M2]**

Simplifying numerator: $3x+3+4x-8-2x=5x-5$ **[M1]**

$\dfrac{5(x-1)}{(x-2)(x+1)}$ **[A1]**

[Total Marks: 7]

20 Factorising numerator to $(x-3)(x+3)$ **[B1]**

Factorising denominator to $(2x-1)(x+3)$ **[B1]**

$\dfrac{x-3}{2x-1}$ **[A1]**

[Total Marks: 3]

21 Denominator of $(x-2)(2x+1)$ **[B1]**

Correct unsimplified numerator: $5(2x+1)-3(x-2)$ **[M1]**

$\dfrac{7x+11}{(x-2)(2x+1)}$ **[A1]**

[Total Marks: 3]

22 Numerator factorised as $(2x-3)(x-3)$ **[B2]**

Allow [M1] for $(2x+a)(x+b)$ where $ab=9$ or $2b+a=-9$

Denominator factorised as $2(x-3)(x+3)$ or $2(x^2-9)$ or $(2x-6)(x+3)$ **[B1]**

$\dfrac{2x-3}{2(x+3)}$ or $\dfrac{2x-3}{2x+6}$ **[A1]**

[Total Marks: 4]

23 $2(x-2)(x+2)$ **[B2]**

If not B2, then B1 for $2(x^2-4)$ or $(2x-4)(x+2)$ **[B1]**

$(x-2)(x+3)$ **[B2]**

If not B2, then B1 for $(x+a)(x+b)$ where $ab=-6$ or $a+b=1$ **[B1]**

$\dfrac{2(x+2)}{x+3}$ or $\dfrac{2x+4}{x+3}$ **[A1]**

[Total Marks: 5]

24 (a) Common denominator $(2x-1)(x+4)$ seen **[M1]**

Numerator: $(x+3)(x+4)-(x-2)(2x-1)$ **[M1]**

$(x^2+7x+12)-(2x^2-5x+2)$ correctly simplified to $-x^2+12x+10$ **[A1]**

(b) Factorising numerator to $(x-4)(x+4)$ or denominator to $x(x+4)$ **[M1]**

$\dfrac{x-4}{x}$ **[A1]**

[Total Marks: 5]

25 Factorising x^2-25 to $(x-5)(x+5)$ and x^2-3x to $x(x-3)$ **[B1]**

Factorising x^2+6x+5 to $(x+1)(x+5)$ and $2x-6$ to $2(x-3)$ **[B1]**

Showing multiplication by the reciprocal, e.g. $\dfrac{(x-5)(x+5)}{x(x-3)} \times \dfrac{2(x-3)}{(x+1)(x+5)}$ **[M1]**

$\dfrac{2(x-5)}{x(x+1)}$ (or equivalent expanded form $\dfrac{2x-10}{x^2+x}$) **[A1]**

[Total Marks: 4]

26 **(a)** M1 for factorising $2x^2 - 8x$ to $2x(x-4)$

M1 for factorising $x^2 - 16$ to $(x-4)(x+4)$

M1 for inverting the second fraction and factorising, i.e., multiplying by $\dfrac{3(x+4)}{6x}$

1 **[A1]**

(b) M1 for writing the denominator $x^2 - 9$ as $(x-3)(x+3)$

M1 for common denominator $(x-3)(x+3)$ and correct numerator $5(x+3) + 2x + 1$

$\dfrac{7x+16}{(x-3)(x+3)}$ or $\dfrac{7x+16}{x^2-9}$ **[A1]**

[Total Marks: 7]

27 **(a)** $\dfrac{4(2x-1) - 3(x-3)}{(x-3)(2x-1)}$ **[M1]**

Numerator expanded correctly to $8x - 4 - 3x + 9$ **[M1]**

$\dfrac{5x+5}{(x-3)(2x-1)}$ or $\dfrac{5(x+1)}{(x-3)(2x-1)}$ **[A1]**

(b) Multiplies by reciprocal: $\times \dfrac{4x+2}{x+5}$ **[M1]**

Factorises $x^2 - 25$ to $(x-5)(x+5)$ or $4x+2$ to $2(2x+1)$ **[M1]**

Factorises denominator $2x^2 - 9x - 5$ to $(2x+1)(x-5)$ **[M1]**

2 **[A1]**

[Total Marks: 7]

28 **(a)** Uses common denominator $(x-2)(x+2)$ or $x^2 - 4$ **[B1]**

Correct unsimplified numerator $3(x+2) - (x+5)$ **[M1]**

$\dfrac{2x+1}{(x-2)(x+2)}$ or $\dfrac{2x+1}{x^2-4}$ **[A1]**

(b) Factorises $4x^2 - 25$ as $(2x-5)(2x+5)$ **[B1]**

Factorises denominators as $x(2x-5)$ and $2x(2x+5)$ **[B2]**

Allow [B1] if only one denominator is factorised correctly

Correct cancellation leading to final answer $3x$ **[A1]**

[Total Marks: 7]

29 **(a)** $\dfrac{5(2x+3) + 4(x-2)}{(x-2)(2x+3)}$ **[M1]**

$10x + 15 + 4x - 8$ **[M1]**

$\frac{14x+7}{(x-2)(2x+3)}$ or $\frac{7(2x+1)}{(x-2)(2x+3)}$ **[A1]**

(b) $\frac{2x^2-5x-12}{4x^2-9} \times \frac{2x-3}{x-4}$ **[M1]**

$(2x+3)(x-4)$ **[B1]**

$(2x-3)(2x+3)$ **[B1]**

1 **[A1]**

[Total Marks: 7]

30 (a) $(2x-1)(x+3)$ seen for numerator **[M1]**

$(2x-1)(2x+1)$ seen for denominator **[M1]**

Identifies and cancels the common factor $(2x-1)$ **[M1]**

$\frac{x+3}{2x+1}$ **[A1]**

(b) Factorises denominator: $x^2-4x+3=(x-1)(x-3)$ **[M1]**

Writes with a common denominator: $\frac{5(x-1)+(x+2)}{(x-1)(x-3)}$ **[M1]**

Simplifies numerator: $5x-5+x+2=6x-3$ **[M1]**

$\frac{3(2x-1)}{(x-1)(x-3)}$ or $\frac{6x-3}{x^2-4x+3}$ **[A1]**

[Total Marks: 8]